Math 162
Riemann vs. Darboux integral

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Theorem 1. *A function $f : [a,b] \to \mathbb{R}$ is Riemann integrable if and only if it is Darboux integrable, in which case the two integrals are equal.*

We will establish the theorem by first proving two lemmas, which were stated without proof in class.

Lemma 2. *Suppose that $f : [a,b] \to \mathbb{R}$ is Darboux integrable. For any $\epsilon > 0$, there exists $\delta > 0$ such that for any partition P of $[a,b]$ satisfying $||P|| < \delta$, we have $U(f,P) - L(f,P) < \epsilon$.*

Proof. Fix $\epsilon > 0$, so by the criterion for Darboux integrability there exists a partition P_0 of $[a,b]$ such that $U(f,P_0) - L(f,P_0) < \frac{\epsilon}{3}$, say $P_0 = \{x_0, x_1, \cdots, x_n\}$. Since f is Darboux integrable it is in particular bounded, so there exists $M > 0$ such that $|f(x)| \leq M$ for all $x \in [a,b]$.

Let $\delta := \min(||P_0||, \frac{\epsilon}{6Mn})$, and suppose that P is a partition of $[a,b]$ satisfying $||P|| < \delta$, written as $P = \{y_0, y_1, \cdots, y_m\}$. Since $||P|| < \delta \leq ||P_0||$, it follows that for any index $1 \leq j \leq m$, there exists at most one $1 \leq i \leq n$ such that $y_{j-1} < x_i < y_j$. Let $J \subset \{1, \cdots, m\}$ be the set of such indices, and note that $\#J \leq n$. Now let $Q := P_0 \cup P$ and consider the difference $U(f,P) - U(f,Q)$. If $1 \leq j \leq m$ is such that $j \notin J$, then y_{j-1}, y_j are consecutive elements of Q, so the corresponding terms in $U(f,P)$ and $U(f,Q)$ cancel out.

Fix $j \in J$, so there exists a unique $1 \leq i \leq n$ such that $y_{j-1} < x_i < y_j$. Let

$$M_j := \sup\{f(x) \mid x \in [y_{j-1}, y_j]\},$$
$$M'_j := \sup\{f(x) \mid x \in [y_{j-1}, x_i]\},$$

and

$$M''_j := \sup\{f(x) \mid x \in [x_i, y_j]\}.$$

Thus

$$U(f,P) - U(f,Q) = \sum_{i \in J} M_j \Delta y_j - M'_j(x_i - y_{j-1}) - M''_j(y_j - x_i),$$

and since

$$\begin{aligned} |M_j \Delta y_j - M'_j(x_i - y_{j-1}) - M''_j(y_j - x_i)| &\leq |M_j|\Delta y_j + |M'_j|(x_i - y_{j-1}) + |M''_j|(y_j - x_i) \\ &\leq M\Delta y_j + M(x_i - y_{j-1}) + M(y_j - x_i) \\ &= 2M\Delta y_j \leq 2M||P|| < 2M\delta, \end{aligned}$$

we obtain

$$U(f,P) - U(f,Q) < \sum_{j \in J} 2M\delta \leq 2Mn\delta < \frac{\epsilon}{3}.$$

Analogously, one shows that $L(f,Q) - L(f,P) < \frac{\epsilon}{3}$. Note that since Q is a refinement of P_0, we have

$$U(f,Q) - L(f,Q) \leq U(f,P_0) - L(f,P_0).$$

Finally, we conclude that

$$\begin{aligned} U(f,P) - L(f,P) &= U(f,P) - U(f,Q) + U(f,Q) - L(f,Q) + L(f,Q) - L(f,P) \\ &\leq U(f,P) - U(f,Q) + U(f,P_0) - L(f,P_0) + L(f,Q) - L(f,P) \\ &< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon. \end{aligned}$$

□

Recall from class that if $f : [a, b] \to \mathbb{R}$ is bounded and (P, τ) is any tagged partition of $[a, b]$, then

$$L(f, P) \le S(f, P, \tau) \le U(f, P).$$

Lemma 3. *Suppose that $f : [a, b] \to \mathbb{R}$ is bounded. For any partition P of $[a, b]$ and any $\epsilon > 0$, there exists taggings τ_1, τ_2 of P such that*

$$U(f, P) - S(f, P, \tau_1) < \epsilon \quad \textit{and} \quad S(f, P, \tau_2) - L(f, P) < \epsilon.$$

Proof. We will prove that there exists a tagging τ of P such that $U(f, P) - S(f, P, \tau) < \epsilon$; the other claim is completely analogous. Writing $P = \{x_0, x_1, \cdots, x_n\}$, for any $1 \le i \le n$ we choose $x_i^* \in [x_{i-1}, x_i]$ so that

$$M_i - f(x_i^*) < \frac{\epsilon}{n||P||},$$

which is possible by the definition of M_i. Then we have

$$U(f, P) - S(f, P, \tau) = \sum_{i=1}^{n} (M_i - f(x_i^*))\Delta x_i \le n(M_i - f(x_i^*))||P|| < \epsilon$$

as needed. □

Proof of Theorem 1. First, assume that f is Riemann integrable with integral I. Fix $\epsilon > 0$, so there exists $\delta > 0$ such that for any tagged partition (P, τ) of $[a, b]$, satisfying $||P|| < \delta$, we have $|S(f, P, \tau) - I| < \frac{\epsilon}{4}$. Choose a partition P such that $||P|| < \delta$ (e.g. an equipartition with n large), and apply Lemma 3 to find taggings τ_1, τ_2 of P such that

$$U(f, P) - S(f, P, \tau_1) < \frac{\epsilon}{4} \quad \text{and} \quad S(f, P, \tau_2) - L(f, P) < \frac{\epsilon}{4}.$$

Then we have

$$\begin{aligned} U(f, P) - L(f, P) &= U(f, P) - S(f, P, \tau_1) + S(f, P, \tau_1) - I + I - S(f, P, \tau_2) + S(f, P, \tau_2) - L(f, P) \\ &\le U(f, P) - S(f, P, \tau_1) + |S(f, P, \tau_1) - I| + |S(f, P, \tau_2) - I| + S(f, P, \tau_2) - L(f, P) \\ &< \frac{\epsilon}{4} + \frac{\epsilon}{4} + \frac{\epsilon}{4} + \frac{\epsilon}{4} = \epsilon, \end{aligned}$$

so f is Darboux integrable as claimed.

For the converse, suppose that f is Darboux integrable with integral I, and fix $\epsilon > 0$. By Lemma 2, there exists $\delta > 0$ such that for any partition P of $[a, b]$ satisfying $||P|| < \delta$, we have $U(f, P) - L(f, P) < \epsilon$. For any tagged partition (P, τ), we have

$$L(f, P) \le S(f, P, \tau) \le U(f, P),$$

and since

$$L(f, P) \le I \le U(f, P)$$

as well, it follows that $||P|| < \delta$ implies that

$$|S(f, P, \tau) - I| < \epsilon$$

as desired. This argument also shows that the Riemann integral of f equals the Darboux integral I, so we are done. □